CYPRESS

EMBEDDED SYSTEMS

Power Monitor
BRN404X Carrier Board

DRAWING NOTES:

Refer to specification 10002-DR-001-00

To be read in conjunction with drawing set 10002-SCH-001-xx LED Driver Board.

REV	REVISION DESCRIPTION	DATE
0	AS BUILT	03/10/2025
1	CORRECTION TO JST PINOUTS TABLE	04/25/2025

Cypress Embedded Systems
1 Cypress Street, Toronto, Ontario

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TITLE:
Power Monitor
BRN404X Carrier Board
Title Page

DRAWN BY: MIKE MCMAHON, LEL

DWG NUMBER: 10002-SCH-002-01 REV 1

DATE: 2025-05-27 2:29 PM

Drawing No.	Drawing Title	Revision
10002-SCH-002-01	TITLE PAGE	1
10002-SCH-002-02	DRAWING LIST	1
10002-SCH-002-03	CARRIER BREAKOUT CIRCUIT	1
10002-SCH-002-04	BILL OF MATERIALS	1

DRAWING NOTES:

Refer to specification 10002-DR-001-00

To be read in conjunction with drawing set 10002-SCH-001-xx LED Driver Board.

REV	REVISION DESCRIPTION	DATE
0	AS BUILT	03/10/2025
1	CORRECTION TO JST PINOUTS TABLE	04/25/2025

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TITLE:

Power Monitor
BRN404X Carrier Board
Drawing List

DRAWN BY: MIKE MCMAHON, LEL

DWG NUMBER: 10002-SCH-002-02 REV 1

DATE: 2025-05-27 2:29 PM

DRAWING NOTES:

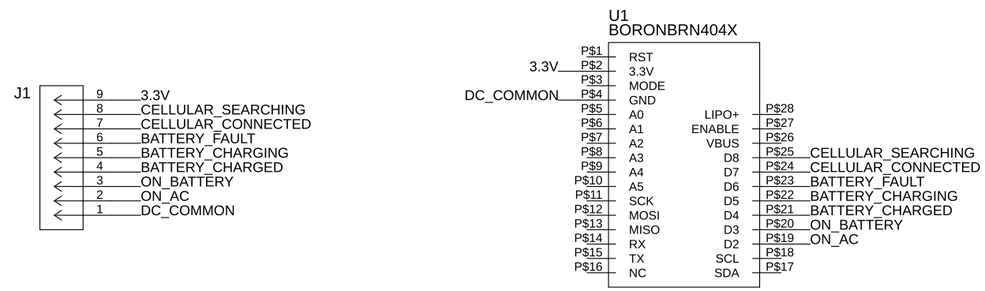
Refer to specification 10002-DR-001-00

To be read in conjunction with drawing set 10002-SCH-001-xx LED Driver Board.

BRN404X CARRIER BREAKOUT CIRCUIT

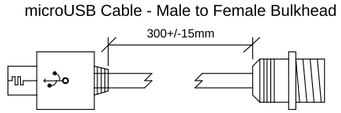
JST Connector pinout:

Pin#	Function
1	DC Common
2	D2 On AC Power
3	D3 On Battery Power
4	D4 Battery Charged
5	D5 Battery Charging
6	D6 Battery Fault
7	D7 Cellular Connected
8	D8 Cellular Searching
9	3.3Vdc



FOR INFORMATION ONLY

microUSB Cable - Male to Female Bulkhead

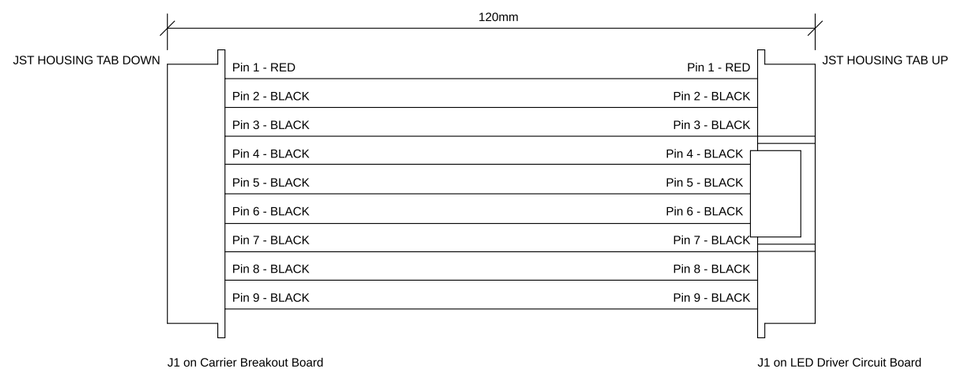


See BOM Item 5.

Threaded end of cable installed on enclosure.
Male end connects to BRN404X.
Center cable jacket and shielding removed to allow for tighter bend radius.

Carrier Breakout to LED Driver Board Interconnect Cable

(For information only. Assembled separately.)



See BOM Items 1, 2, 3, 6, 7, and 8.

REV	REVISION DESCRIPTION	DATE
0	AS BUILT	03/10/2025
1	CORRECTION TO JST PINOUTS TABLE	04/25/2025

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TITLE:

Power Monitor
BRN404X Carrier Board
Carrier Breakout Circuit

DRAWN BY: MIKE MCMAHON, LEL

DWG NUMBER: 10002-SCH-002-03 REV 1

DATE: 2025-05-27 2:29 PM

Item	Qty	Reference	Manufacturer	Part Number	Description
1	1	J1	JST SALES AMERICA	B09B-PASK-1	CONN HEADER VERT 9POS 2MM
2	1	J1	JST SALES AMERICA	PAP-09V-S	CONN RCPT HSG 9POS 2.00MM
3	9	J1	JST SALES AMERICA	SPHD-002T-P0.5	CONN SOCKET 24-28AWG CRIMP TIN
4	1	U1	PARTICLE	BRN404X	PARTICLE BORON BRN404X LTE SOM
5	1	USB BULKHEAD CONNECTOR CABLE	PREMIER CABLE	P6068	ADAFRUIT 6068 MICRO USB BULKHEAD CONNECTOR
6	12cm	INTERCONNECT CABLE	CNC Tech	10064-24-1-0500-004-1-TS	HOOK-UP STRND 24AWG 30V RED FEET
7	96cm	INTERCONNECT CABLE	CNC Tech	10064-24-1-0500-001-1-TS	HOOK-UP STRND 24AWG 30V BLK FEET
8	8cm	INTERCONNECT CABLE	FITCO nv	0508000002	PROTEC WRAP 5MM FR BLACK METER

DRAWING NOTES:

Refer to specification 10002-DR-001-00

To be read in conjunction with drawing set 10002-SCH-001-xx LED Driver Board.

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TITLE:

Power Monitor
BRN404X Carrier Board
Bill of Material

DRAWN BY: MIKE MCMAHON, LEL

DWG NUMBER: 10002-SCH-002-04 REV 1

DATE: 2025-05-27 2:29 PM